

Q. 1 (Event gymnastics). We are given three events E_1, E_2, E_3 .

- a. The event that both E_1 and E_2 occur, but E_3 does not occur is

$$E_1 \cap E_2 \cap E_3^c.$$

- b. The event that exactly two of the three events occur is

$$(E_1 \cap E_2 \cap E_3^c) \cup (E_1 \cap E_2^c \cap E_3) \cup (E_1^c \cap E_2 \cap E_3).$$

or

$$((E_1 \cap E_2) \cup (E_1 \cap E_3) \cup (E_2 \cap E_3)) \cap (E_1 \cap E_2 \cap E_3)^c$$

- c. The event that at least one event occurs is

$$E_1 \cup E_2 \cup E_3.$$

- d. The event that none of the three events occur is

$$E_1^c \cap E_2^c \cap E_3^c = (E_1 \cup E_2 \cup E_3)^c.$$

- e. $E_1 \triangle E_2 = (E_1 \setminus E_2) \cup (E_2 \setminus E_1) = (E_1 \cup E_2) \cap (E_1 \cap E_2)^c = (E_1 \cap E_2^c) \cup (E_1^c \cap E_2)$ is the event that exactly one of E_1 and E_2 occurs.

Q. 2 (Coin flips). We consider three coin flips first.

- a. Let us denote *heads* by H and *tails* by T . We may represent each possible outcome by a three-letter word formed of H and T . For instance, HTT denotes the outcome that the first flip comes up H and the others come up T . Hence, a good sample space for this model is

$$\Omega = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}.$$

Since we assume that each possible outcome is equally likely to occur, the probability measure $\mathbb{P}$ must satisfy

$$\mathbb{P}\{HHH\} = \dots = \mathbb{P}\{TTT\}.$$

Note that $\{HHH\}, \dots, \{TTT\}$ are mutually exclusive events and their union gives Ω . By the properties of probability measure,

$$\mathbb{P}\{HHH\} + \dots + \mathbb{P}\{TTT\} = \mathbb{P}(\Omega) = 1.$$

Hence, $\mathbb{P}\{HHH\} = \dots = \mathbb{P}\{TTT\} = \frac{1}{|\Omega|} = \frac{1}{8}$. Finally, for each event A ,

$$\mathbb{P}(A) = \mathbb{P}\left(\bigcup_{\omega \in A} \{\omega\}\right) = \sum_{\omega \in A} \mathbb{P}\{\omega\} = \frac{|A|}{8},$$

which defines the probability measure $\mathbb{P}$.

- b. The event that the first coin comes up heads is

$$E_1 = \{HHH, HHT, HTH, HTT\},$$

and its probability is $\mathbb{P}(E_1) = \frac{|E_1|}{8} = \frac{4}{8} = \frac{1}{2}$. The event that the second coin comes up tails is

$$E_2 = \{HTH, HTT, TTH, TTT\},$$

and its probability is $\mathbb{P}(E_2) = \frac{1}{2}$.

- c. The event that an odd number of coins come up heads is

$$E_3 = \{HHH, HTT, THT, TTH\},$$

and the probability is $\mathbb{P}(E_3) = \frac{1}{2}$.

Now we consider 100 coin flips.

- d. Let us denote by $\omega = (\omega_1, \dots, \omega_{100})$ the outcome that the first flip comes up $\omega_1 \in \{H, T\}$, ..., the 100th flip comes up $\omega_{100} \in \{H, T\}$. Hence, a good sample space for this model is

$$\Omega = \{\omega = (\omega_1, \dots, \omega_{100}) : \omega_i \in \{H, T\} \text{ for every } 1 \leq i \leq 100\}.$$

Note that $|\Omega| = 2^{100}$ since there are two possible values of ω_i for each $1 \leq i \leq 100$.

The event that the n th coin comes up heads is

$$\{\omega \in \Omega : \omega_n = H\}.$$

Since we assume that each outcome is equally likely to occur, the probability measure $\mathbb{P}$ satisfies

$$\mathbb{P}\{\omega\} = \frac{1}{|\Omega|} = \frac{1}{2^{100}} \text{ for each } \omega \in \Omega,$$

which follows by the same reasoning as in part a. Finally, for each event A ,

$$\mathbb{P}(A) = \sum_{\omega \in A} \mathbb{P}\{\omega\} = \frac{|A|}{2^{100}},$$

which defines the probability measure $\mathbb{P}$.

- e. Let us denote by A the event that exactly one coins come up heads. Note that there are 100 different sequences of coin flips where exactly one flip comes up heads. Hence, $|A| = 100$ and

$$\mathbb{P}(A) = \frac{|A|}{2^{100}} = \frac{100}{2^{100}} \approx 7.9 \times 10^{-29}.$$

That is, the probability of observing exactly one heads in a sequence of 100 flips is virtually zero.

Now we consider an *infinite* sequence of coin flips.

- f. Let us denote by $\omega = (\omega_1, \omega_2, \dots)$ the outcome that the first flip comes up $\omega_1 \in \{H, T\}$, the second flip comes up $\omega_2 \in \{H, T\}$, ... Hence, a good sample space for this model is

$$\Omega = \{\omega = (\omega_1, \omega_2, \dots) : \omega_i \in \{H, T\} \text{ for every } i \geq 1\}.$$

- g. Note that the sample space Ω in part f. has infinitely (indeed uncountably¹) many outcomes. Hence, any probability measure $\mathbb{P}$ must assign zero probability to every outcome, that is, $\mathbb{P}\{\omega\} = 0$ for every $\omega \in \Omega$. Since Ω is an uncountable set, knowing

¹Here is one way of arguing that Ω is uncountable. Note that every real number $r \in [0, 1]$ has a binary expansion of the form $0.r_1r_2\dots$ where $r_i \in \{0, 1\}$ for each $i \geq 1$. (If the binary expansion $0.r_1r_2\dots$ of a number r is given, then the number can be computed by the formula $r = \sum_{i=1}^{\infty} r_i 2^{-i}$.) For instance $0 = 0.00\dots$, $0.5 = 0.100\dots$, $1 = 0.11\dots$. For every decimal digit r_i , we may interpret $r_i = 0$ as getting T and $r_i = 1$ as getting H for the i th coin flip in our random experiment. This way, every number $r \in [0, 1]$ corresponds to an infinite sequence of coin flips, that is, an outcome $\omega \in \Omega$. Symmetrically, every outcome ω can be represented by a number $r \in [0, 1]$. This forms a one-to-one correspondence between the sets Ω and $[0, 1]$. It is well-known that $[0, 1]$ is uncountable. Hence, Ω is also uncountable.

the probabilities of individual outcomes does not help in determining the probability $\mathbb{P}(A)$ of an arbitrary event A : for an event A with uncountably many outcomes, the equality in

$$\mathbb{P}(A) \stackrel{?}{=} \sum_{\omega \in A} \mathbb{P}\{\omega\}$$

fails to hold in general (for instance, for $A = \Omega$, which has $\mathbb{P}(\Omega) = 1$). However, it is difficult but possible to construct a probability measure on Ω .

In the lectures, we have already seen an example of a probability measure on an uncountable sample space. In the random experiment of waiting for a bus (Example 1.16 in the notes), the sample space is $\Omega = [0, 1]$ and the probability $\mathbb{P}(A)$ of an event A is defined as the length of A . Since a set with a single point has zero length, we have $\mathbb{P}\{\omega\} = 0$ for every $\omega \in \Omega$.

- Q. 3 (Darts).** a. Since the random point (outcome) is picked from the disc D (we assume that we can always hit the dartboard), a natural sample space for this random experiment is $\Omega = D$.
- b. Let us try to define the probability of hitting a subset $A \subseteq D$ of the dartboard to be proportional to the area of A , that is,

$$\mathbb{P}(A) = k \cdot \text{area}(A)$$

for some $k \in \mathbb{R}$ (which is the same number for every event A).

In order for $\mathbb{P}$ to make sense, we need to check that it satisfies the definition of probability measure. First,

$$1 = \mathbb{P}(\Omega) = k \cdot \text{area}(\Omega) = k \cdot \pi 1^2 = k\pi,$$

hence we must choose $k = \frac{1}{\pi}$. It is also clear that $0 \leq \mathbb{P}(A) \leq 1$ for every event $A \subset \Omega$. Finally, for disjoint events A, B ,

$$\mathbb{P}(A \cup B) = \frac{\text{area}(A \cup B)}{\pi} = \frac{\text{area}(A) + \text{area}(B)}{\pi} = \mathbb{P}(A) + \mathbb{P}(B),$$

and we can check that the similar property holds for a sequence $E_1, E_2, \dots$ of disjoint events.

Therefore, the assignment

$$\mathbb{P}(A) = \frac{\text{area}(A)}{\pi}$$

to an arbitrary event A defines a probability measure $\mathbb{P}$.

- c. We have

$$\mathbb{P}(R) = \frac{\text{area}(R)}{\pi} = \frac{\pi \cdot \frac{3}{4} - \pi \cdot \frac{1}{2}}{\pi} = \frac{3}{4} - \frac{1}{2} = \frac{1}{4} = 0.25.$$

- d. Let us first represent the sample space $\Omega = D$ in polar coordinates. Note that D is precisely the set of points on the plane whose distance r from the center satisfies $r \leq 1$ (and the angle ϕ can take any value in $[0, 2\pi[$). In polar coordinates notation,

$$\Omega = \{(r, \phi) \in [0, +\infty[\times [0, 2\pi[: r \leq 1\} = [0, 1] \times [0, 2\pi[.$$

Similarly, the ring R is precisely the set of points on the plane whose distance r from the center satisfies $\frac{1}{\sqrt{2}} < r < \frac{\sqrt{3}}{2}$, that is,

$$R = \left\{ (r, \phi) \in \Omega : \frac{1}{\sqrt{2}} < r < \frac{\sqrt{3}}{2} \right\}$$

Let us denote by $\tilde{\mathbb{P}}$ the probability measure under the new modelling assumptions. In this model,

$$\tilde{\mathbb{P}}(R) = \text{length} \left(\left[\frac{1}{\sqrt{2}}, \frac{\sqrt{3}}{2} \right] \right) = \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} = \frac{\sqrt{3} - \sqrt{2}}{2} \approx 0.16.$$

(Here, the proportionality constant is 1 as $\text{length}([0, 1]) = 1$.)

- e. The answers to parts c. and d. are different. Working under different modelling assumptions in parts c. and d. leads to different probability measures $\mathbb{P}$ and $\tilde{\mathbb{P}}$ on the same sample space $\Omega = D$. In particular, the probability measure in part c represents a uniform scattering of points across the disc, whereas the measure in part d represents more of a clustering phenomenon around the center.